DESIGN BASELINE EVALUATION

Design baseline

Revision Aluminum box drum arms 2026-10-24 · plate-frame frame · x44-standard motors

Named input case · design_baseline.json

2026-10-24T05:34:52+00:00

DEFAULT rigid-wheel BWR: published loose GRC-1 parameters and equal wheel loads. Student-selected conservative design assumption; not prototype calibrated. Point turns retain their existing model. Separate empty/full BWR slope requirements check slip<1.

6 modeled checks need attention

59 / 65 physical model checks pass · 6 capabilities not assessed

Predictions are conditional on the failed checks below. These are predictions from the stated inputs, not measured performance.

Dry rover mass	Payload per cycle	Complete cycles	Delivered volume
52.7 kg	**116.1 kg**	**5**	**348.4 L**
Before adding regolith	Total carried by four drums	Within 15 minutes; no partial delivery credit	Complete-cycle mission estimate

Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

Construction points	Mass points	Energy points
3,230.49	**2,187.18**	**1,043.31**
Weighted mass + energy points	Dry-mass productivity component	Consumed-energy productivity component

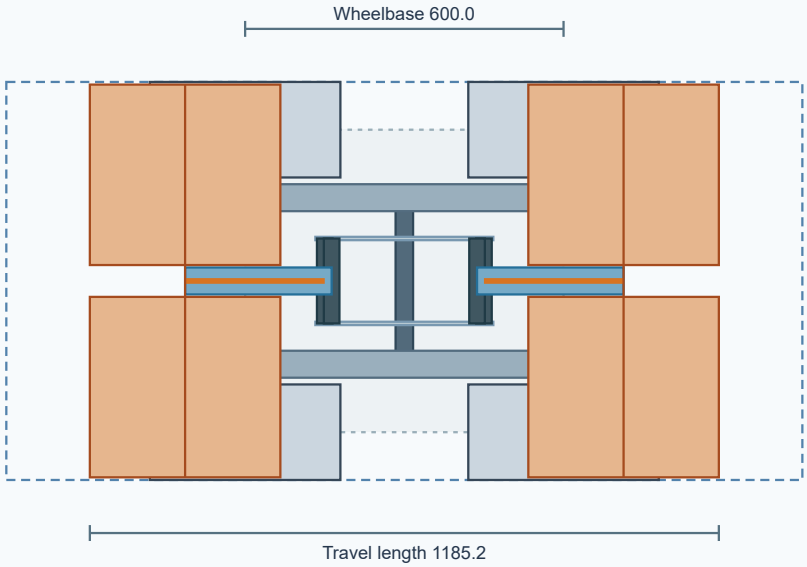
01 / ROVER CONFIGURATION

Geometry at the evaluated inputs

Level-ground schematics and one prescribed obstacle pose. Check statuses below determine whether envelopes and clearance requirements are satisfied.

Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.



Wheel-center track: 570.0 | Overall width: 750.0 | Wheel width: 180.0

Each drum: 359.0 diameter x 340.0 axial length | Pair gap: 60.0

Blue: one arm per drum pair · orange: actuators · charcoal: CF connectors · slate: rockers and axle.

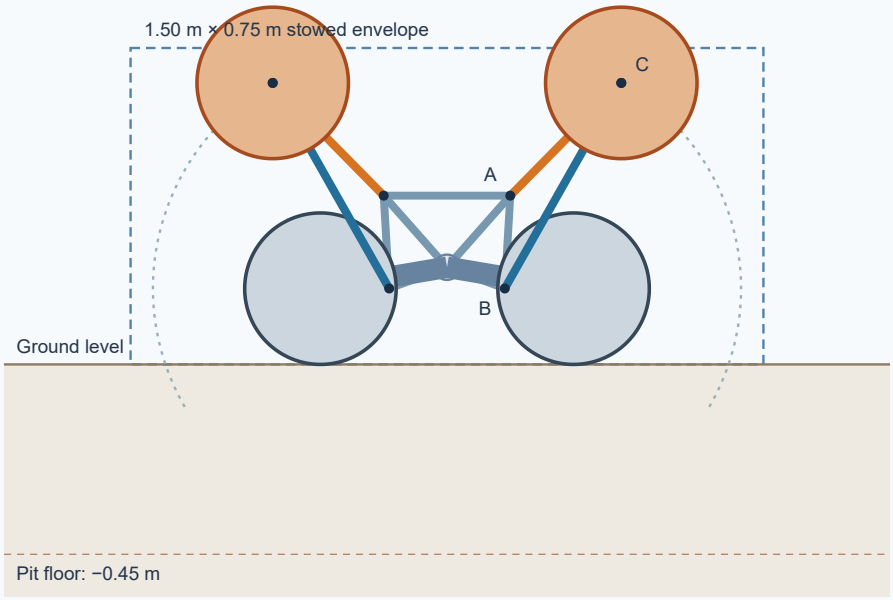
Four CF tubes: upper/lower footprints can overlap. Dashed blue: 1500 x 750 stowed limit.

Plate outlines are schematic. Projection overlap does not prove collision. Heights and equipment are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

Raised · travel / deposition

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



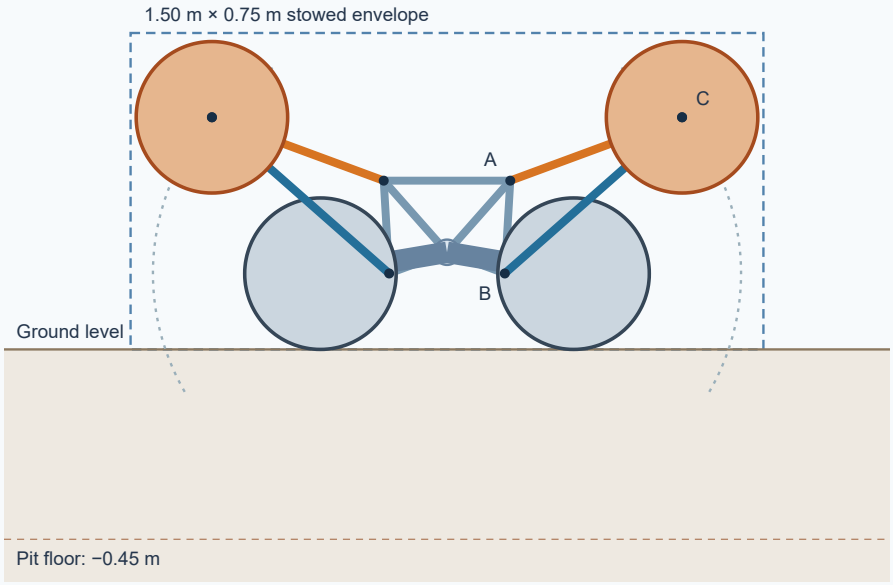
Actuator: 375.0 mm · Fixed arm: 560.0 mm · Drum Ø 359.0 mm

Drum top: +0.847 m · Drum bottom: +0.488 m · A: 0.400 m · B: 0.180 m

Blue = fixed arm · Orange = actuator · Attachment B-D: 560.0 mm (100.0% of arm) · Ground is y = 0

Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



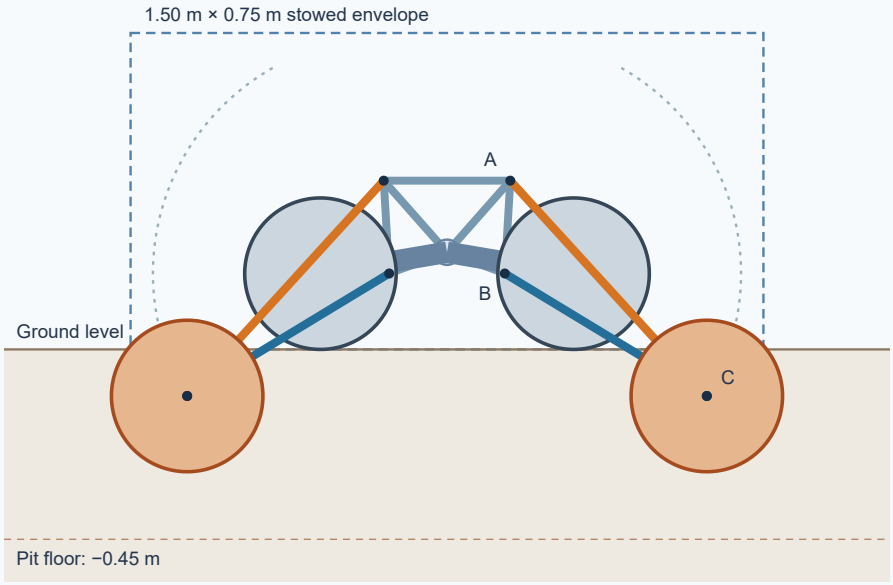
Actuator: 434.0 mm · Fixed arm: 560.0 mm · Drum Ø 359.0 mm

Drum top: +0.730 m · Drum bottom: +0.371 m · A: 0.400 m · B: 0.180 m

Blue = fixed arm · Orange = actuator · Attachment B-D: 560.0 mm (100.0% of arm) · Ground is y = 0

Extended · digging

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



Actuator: 691.0 mm · Fixed arm: 560.0 mm · Drum Ø 359.0 mm

Drum top: +0.069 m · Drum bottom: -0.290 m · A: 0.400 m · B: 0.180 m

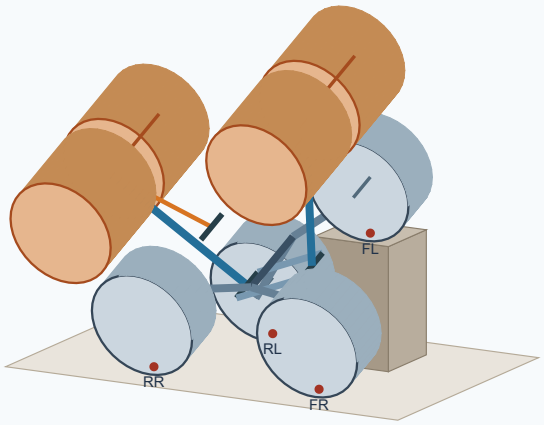
Blue = fixed arm · Orange = actuator · Attachment B-D: 560.0 mm (100.0% of arm) · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

One wheel on the rock / travel pose

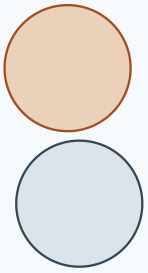
Front-left raised: 400.0 mm · chassis pitch 22.4° · roll 19.0°

Same kinematic pose as the model; arms held in travel position.



Common roll removed

Closest cylinder pair: drum 1 / FL



Radial circle gap: +28.32 mm

Axial end gap: -175.00 mm

Finite-cylinder gap: +28.32 mm

Required clearance: 15.00 mm

One centered arm/actuator per drum pair · blue: arms · orange: actuators · charcoal: CF rods · gray: wheels · tan: drums.

Support block shows the prescribed height; its shape and width are illustrative. Actual rock climbing is not simulated.

Contact model omits camber effects. The right section also needs the axial-width check; circles alone can over-reject.

Parallel-cylinder clearance is the default model. This section shows the closest pair for the illustrated front-left raise.

Plate-frame member schematic

Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.

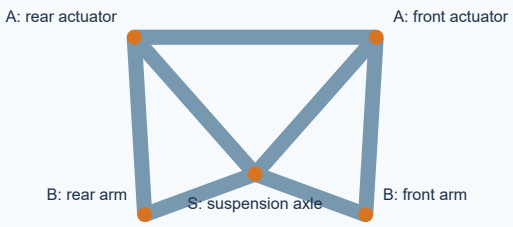


Plate thickness: 6.35 mm (fixed)

Common strip width: 18.75 mm

CF ID / OD: 25.40 / 29.40 mm

Plate spacing / CF span: 160.0 mm

Rocker bearing spacing: 313.8 mm

Wheel-to-rocker gap: 12.7 mm

Aluminum arm box: 50.80 x 50.80 mm

Arm wall: 1.5875 mm

Central support: pin, balanced mirrored gravity only.

CF material properties are PROVISIONAL.

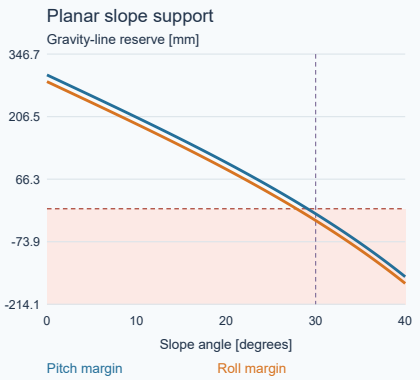
Spoke buckling is a strip screen,

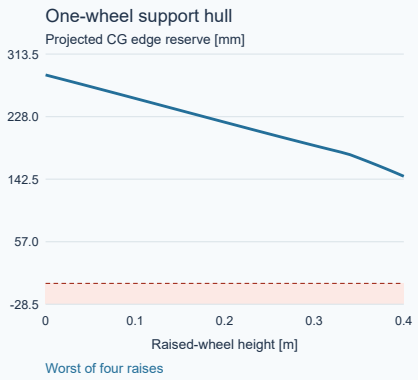
not whole-plate buckling validation.

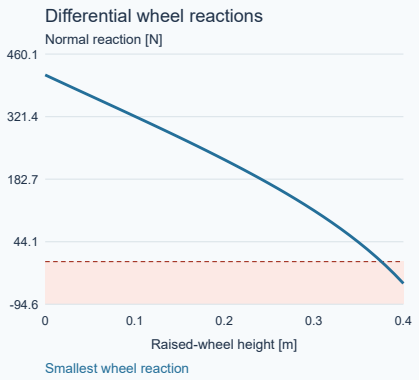
Static slope and articulation curves

Loaded travel pose: gravity stability screens

One wheel raised at a time on horizontal platforms. Curves are model calculations; no physical rock profile is simulated.







02 / PREDICTED OUTPUT

Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

Prediction	Value	Meaning
Full cycle duration	159.6 s	Includes digging, actuation, travel, dumping and relocation; overlap removed once.
Energy per cycle	6.68 Wh	Electrical energy including auxiliaries.
Mission energy · complete cycles	34.1 Wh	Includes idle electronics for remaining mission time.
Mission energy · continuous estimate	37.6 Wh	Used by the existing battery constraint and MDO objective.
Peak electrical power	1,205 W	Largest modeled simultaneous demand; not an average power rating.
Loaded mass	168.8 kg	Dry mass + all four drums' regolith payload.
Actual route peak speed	0.57 m/s	Prescribed motion, not measured vehicle performance.
Score · complete-cycle estimate	2,961.45 points	Uses complete deliveries and idle remainder; separate from the continuous MDO objective.
Empty rover · 30° climb	No climbing solution	BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 38.02 N. Soil-fit range flags: 0. Acceptance: solved slip strictly below 1.
Full rover · 20° climb	No climbing solution	BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 47.64 N. Soil-fit range flags: 1. Acceptance: solved slip strictly below 1.

Dry mass breakdown

Each assembly is counted once.

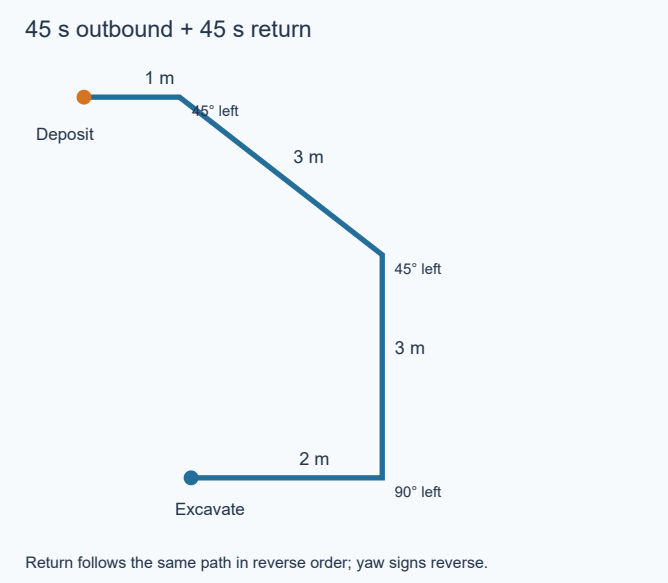
Equipment allowance	6.00 kg
Frame	5.34 kg
Four bare wheels	13.0 kg
Four drive assemblies	4.15 kg
Four drums + two drive assemblies	13.9 kg
Battery	1.85 kg
Arms + actuators	8.46 kg

Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

Digging	2.89 Wh
Dumping	0.15 Wh
Travel + deposition turn	1.96 Wh
Actuators	0.48 Wh
Fresh-site move	0.08 Wh
Electronics	1.11 Wh

Route & cycle accounting



45 s outbound + 45 s return

Dig and lower concurrently → raise loaded drums → outbound → dump one pair → turn → dump second pair → return → relocate to fresh ground.

Lowering overlap is subtracted once. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 159.6 s.

03 / GAPS TO RESOLVE

Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Empty rover: 30° BWR climb `empty_slope_traversal_margin`	No climbing solution `Slip = 1 is a failure sentinel`	Slip < 1 (strict)	-0.34	Fail
Full rover: 20° BWR climb `loaded_slope_traversal_margin`	No climbing solution `Slip = 1 is a failure sentinel`	Slip < 1 (strict)	-0.12	Fail
Turning: traction reserve `turn_traction_margin`	Reserve: -112.6 N	Reserve ≥ 0	-112.6 N	Fail
Asymmetric dump tip reserve `tip_stability_margin`	Reserve: -22.9 mm	Reserve ≥ 0	-22.9 mm	Fail
30° slope: static tip reserve `slope_tip_margin`	Reserve: -26.2 mm	Reserve ≥ 0	-26.2 mm	Fail
Rock path: smallest wheel reaction `rock_sweep_wheel_load_margin`	Reserve: -48.4 N	Reserve ≥ 0	-48.4 N	Fail
Dry rover mass `dry_mass`	52.7 kg	≤ 80.0 kg	27.3 kg	Pass
Compact overall length `rover_length`	1,473 mm	≤ 1,500 mm	26.6 mm	Pass
Overall width `rover_width`	750.0 mm	≤ 750.0 mm	0.00 mm	Pass
Compact overall height `rover_height`	729.7 mm	≤ 750.0 mm	20.3 mm	Pass
Excavation shaft-power reserve `drum_power_margin`	Reserve: 710.1 W	Reserve ≥ 0	710.1 W	Pass
Excavation: penetration reserve `penetration_margin`	Reserve: 490.9 N	Reserve ≥ 0	490.9 N	Pass
Excavation: traction reserve `dig_traction_margin`	Reserve: 286.8 N	Reserve ≥ 0	286.8 N	Pass
Cut depth margin `cut_depth_margin`	Reserve: 3.50 mm	Reserve ≥ 0	3.50 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Ground contact pressure `ground_pressure_margin`	17.0 kPa	≤ 25.0 kPa	7.99 kPa	Pass
Straight travel: motor-force reserve `straight_motor_margin`	Reserve: 3,503 N	Reserve ≥ 0	3,503 N	Pass
Turning: motor-force reserve `turn_motor_margin`	Reserve: 5,282 N	Reserve ≥ 0	5,282 N	Pass
Straight travel: traction reserve `straight_traction_margin`	Reserve: 238.5 N	Reserve ≥ 0	238.5 N	Pass
Straight travel: shaft-power reserve `straight_power_margin`	Reserve: 2,460 W	Reserve ≥ 0	2,460 W	Pass
Turning: shaft-power reserve `turn_power_margin`	Reserve: 2,571 W	Reserve ≥ 0	2,571 W	Pass
Battery usable energy `battery_energy_margin`	37.6 Wh	≤ 207.4 Wh	169.7 Wh	Pass
Battery peak-power capacity `battery_power_margin`	1,205 W	≤ 1,728 W	523.4 W	Pass
Time remaining after one complete cycle `complete_cycle_margin`	Reserve: 740.4 s	Reserve ≥ 0	740.4 s	Pass
Minimum digging-depth reserve `dig_reach_margin`	Reserve: 284.9 mm	Reserve ≥ 0	284.9 mm	Pass
Pit depth margin `pit_depth_margin`	Reserve: 160.1 mm	Reserve ≥ 0	160.1 mm	Pass
Transport clearance margin `transport_clearance_margin`	Reserve: 290.7 mm	Reserve ≥ 0	290.7 mm	Pass
Travel drum height above wheel-diameter step `wheel_first_margin`	Reserve: 128.7 mm	Reserve ≥ 0	128.7 mm	Pass
Startup position margin `startup_position_margin`	Reserve: 59.0 mm	Reserve ≥ 0	59.0 mm	Pass
Dig position margin `dig_position_margin`	Reserve: 257.0 mm	Reserve ≥ 0	257.0 mm	Pass
Pivot spacing margin `pivot_spacing_margin`	Reserve: 140.4 mm	Reserve ≥ 0	140.4 mm	Pass
Pivot mount margin `pivot_mount_margin`	Reserve: 309.0 mm	Reserve ≥ 0	309.0 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Pivot longitudinal margin `pivot_longitudinal_margin`	Reserve: 163.0 mm	Reserve ≥ 0	163.0 mm	Pass
Actuator pivot longitudinal margin `actuator_pivot_longitudinal_margin`	Reserve: 150.0 mm	Reserve ≥ 0	150.0 mm	Pass
Bottom mount margin `bottom_mount_margin`	Reserve: 0.50 mm	Reserve ≥ 0	0.50 mm	Pass
Stroke margin `stroke_margin`	Reserve: 276.0 mm	Reserve ≥ 0	276.0 mm	Pass
Actuator body margin `actuator_body_margin`	Reserve: 19.0 mm	Reserve ≥ 0	19.0 mm	Pass
Triangle retracted margin `triangle_retracted_margin`	Reserve: 20.4 mm	Reserve ≥ 0	20.4 mm	Pass
Triangle extended margin `triangle_extended_margin`	Reserve: 74.4 mm	Reserve ≥ 0	74.4 mm	Pass
Toggle margin `toggle_margin`	Reserve: 261.1 mm	Reserve ≥ 0	261.1 mm	Pass
Actuator toggle margin `actuator_toggle_margin`	Reserve: 231.9 mm	Reserve ≥ 0	231.9 mm	Pass
Wheel clearance margin `wheel_clearance_margin`	Reserve: 17.4 mm	Reserve ≥ 0	17.4 mm	Pass
Deployed height margin `deployed_height_margin`	Reserve: 1,653 mm	Reserve ≥ 0	1,653 mm	Pass
Actuator force capacity `actuator_force_margin`	2,243 N	≤ 2,333 N	89.9 N	Pass
Excavation torque reserve `drum_torque_margin`	Reserve: 50.5 N·m	Reserve ≥ 0	50.5 N·m	Pass
Dumping torque reserve `dump_torque_margin`	Reserve: 71.7 N·m	Reserve ≥ 0	71.7 N·m	Pass
Minimum payload reserve `payload_margin`	Reserve: 116.1 kg	Reserve ≥ 0	116.1 kg	Pass
Excavated trench diameter reserve `trench_traverse_margin`	Reserve: 69.1 mm	Reserve ≥ 0	69.1 mm	Pass
Fresh site margin `fresh_site_margin`	Reserve: 60.0 mm	Reserve ≥ 0	60.0 mm	Pass
Frame section margin `frame_section_margin`	Reserve: 7.25 mm	Reserve ≥ 0	7.25 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Arm section margin `arm_section_margin`	Reserve: 11.1 mm	Reserve ≥ 0	11.1 mm	Pass
Structural stress envelope `structure_stress_margin`	109.2 MPa	≤ 120.0 MPa	10.8 MPa	Pass
Structural deflection envelope `structure_deflection_margin`	0.60 mm	≤ 5.00 mm	4.40 mm	Pass
Structural buckling reserve `structure_buckling_margin`	Reserve: 1,436 N	Reserve ≥ 0	1,436 N	Pass
Level-route dynamic tip reserve `mobility_stability_margin`	Reserve: 266.5 mm	Reserve ≥ 0	266.5 mm	Pass
Rock pose: geometric reach reserve `obstacle_pose_margin`	Reserve: 149.8 mm	Reserve ≥ 0	149.8 mm	Pass
Rock endpoint: global tip reserve `obstacle_tip_margin`	Reserve: 146.4 mm	Reserve ≥ 0	146.4 mm	Pass
Obstacle drum–wheel clearance `obstacle_drum_wheel_margin`	28.3 mm	≥ 15.0 mm	13.3 mm	Pass
Rock path: global tip reserve `rock_sweep_tip_margin`	Reserve: 146.4 mm	Reserve ≥ 0	146.4 mm	Pass
Available excavation area `excavation_area_margin`	Reserve: 5.36 m^2	Reserve ≥ 0	5.36 m^2	Pass
Fresh-site move: drive/traction reserve `site_drive_margin`	Reserve: 202.9 N	Reserve ≥ 0	202.9 N	Pass
Plate geometry reserve `plate_geometry_margin`	Reserve: 37.6 mm	Reserve ≥ 0	37.6 mm	Pass
Frame packaging reserve `frame_packing_margin`	Reserve: 63.7 mm	Reserve ≥ 0	63.7 mm	Pass
Plate strip stress `plate_stress_margin`	16.6 MPa	≤ 120.0 MPa	103.4 MPa	Pass
Plate strip buckling reserve `plate_buckling_margin`	Reserve: 1,436 N	Reserve ≥ 0	1,436 N	Pass
CF tube bending stress `cf_bending_margin`	105.4 MPa	≤ 175.0 MPa	69.6 MPa	Pass
Wheel diameter search range `wheel_diameter_bounds`	359.0 mm	180.0 mm to 359.0 mm	0.00 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Wheel width search range `wheel_width_bounds`	180.0 mm	50.0 mm to 359.0 mm	130.0 mm	Pass
Drum outer diameter search range `drum_diameter_bounds`	359.0 mm	160.0 mm to 359.0 mm	0.00 mm	Pass
Width of each drum search range `drum_length_bounds`	340.0 mm	120.0 mm to 359.0 mm	19.0 mm	Pass
Excavation speed search range `drum_rpm_bounds`	120.0 rpm	5.00 rpm to 120.0 rpm	0.00 rpm	Pass
Dumping speed search range `dump_rpm_bounds`	60.0 rpm	5.00 rpm to 60.0 rpm	0.00 rpm	Pass
Drum reduction ratio search range `drum_gear_ratio_bounds`	40.0	9.00 to 125.0	31.0	Pass
Scoop protrusion / cut depth search range `cut_depth_bounds`	14.0 mm	3.00 mm to 25.0 mm	11.0 mm	Pass
Arm wall thickness search range `arm_wall_bounds`	1.59 mm	1.00 mm to 5.00 mm	0.59 mm	Pass
Battery capacity search range `battery_capacity_bounds`	259.2 Wh	10.0 Wh to 300.0 Wh	40.8 Wh	Pass
Actuator speed search range `actuator_speed_bounds`	0.07 m/s	0.005 m/s to 0.08 m/s	0.01 m/s	Pass
Travel actuator length search range `travel_actuator_length_bounds`	375.0 mm	300.0 mm to 650.0 mm	75.0 mm	Pass
Compact actuator length search range `actuator_retracted_length_bounds`	434.0 mm	250.0 mm to 650.0 mm	184.0 mm	Pass
Digging actuator length search range `actuator_extended_length_bounds`	691.0 mm	450.0 mm to 1,050 mm	241.0 mm	Pass
Actuator pivot height search range `top_pivot_height_bounds`	400.0 mm	350.0 mm to 700.0 mm	50.0 mm	Pass
Actuator pivot longitudinal position search range `actuator_pivot_x_bounds`	150.0 mm	50.0 mm to 300.0 mm	100.0 mm	Pass
Arm pivot height search range `bottom_pivot_height_bounds`	180.0 mm	90.0 mm to 300.0 mm	90.0 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Arm length search range `arm_length_bounds`	560.0 mm	300.0 mm to 700.0 mm	140.0 mm	Pass
Arm pivot longitudinal position search range `pivot_x_bounds`	137.0 mm	50.0 mm to 300.0 mm	87.0 mm	Pass
Wheelbase search range `wheelbase_bounds`	600.0 mm	380.0 mm to 650.0 mm	50.0 mm	Pass
Wheel-center track search range `track_width_bounds`	570.0 mm	500.0 mm to 699.0 mm	70.0 mm	Pass
Actuator rating, each search range `actuator_force_rating_bounds`	3,500 N	500.0 N to 6,000 N	2,500 N	Pass
Aluminum arm box width search range `arm_box_width_bounds`	50.8 mm	25.0 mm to 100.0 mm	25.8 mm	Pass
Aluminum arm box height search range `arm_box_height_bounds`	50.8 mm	25.0 mm to 100.0 mm	25.8 mm	Pass
CF tube span / plate spacing search range `cf_tube_span_bounds`	160.0 mm	160.0 mm to 460.0 mm	0.00 mm	Pass
CF wall thickness search range `cf_tube_wall_bounds`	2.00 mm	1.00 mm to 6.00 mm	1.00 mm	Pass
Plate strip width search range `plate_spoke_width_bounds`	18.8 mm	12.0 mm to 70.0 mm	6.75 mm	Pass
Rocker box width search range `rocker_box_width_bounds`	50.8 mm	20.0 mm to 65.0 mm	14.2 mm	Pass
Rocker box height search range `rocker_box_height_bounds`	50.8 mm	25.0 mm to 90.0 mm	25.8 mm	Pass
Rocker wall search range `rocker_wall_bounds`	1.59 mm	1.00 mm to 5.00 mm	0.59 mm	Pass
Axle outer diameter search range `axle_outer_diameter_bounds`	33.0 mm	25.0 mm to 90.0 mm	8.00 mm	Pass
Axle wall search range `axle_wall_bounds`	1.00 mm	1.00 mm to 6.00 mm	0.00 mm	Pass
Rocker pivot above wheel centers search range `rocker_pivot_offset_bounds`	50.8 mm	20.0 mm to 150.0 mm	30.8 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Slope motor limits / sideways sliding	The separate BWR climb checks accept slip<1 only. Motor capability on slopes, lateral sliding and controller behavior are not evaluated.	—	—	Not assessed
Real 40 cm rock traversal	Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence.	—	—	Not assessed
50 cm ditch crossing	Ditch width/radius and contact sequence have not been defined.	—	—	Not assessed
Continuous motor operation	Stator-current limits and thermal duty remain undecided.	—	—	Not assessed
Connections & material validation	Bonded inserts, bushings, bolts, local failure and selected CF tube properties need separate checks.	—	—	Not assessed
Soil and skid-steer predictions	Rolling, traction, capture and scrub relationships remain uncalibrated.	—	—	Not assessed

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

04 / ENGINEERING LOADS

Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

Quantity / scope	Prediction	Load case / interpretation
Required force per actuator `required_actuator_force`	2,243 N	Loaded raising/extraction envelope, including the model's breakout allowance.
Frame gravity reaction at each actuator `frame_gravity_actuator_force`	2,910 N	Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load.
Nominal force per wheel `frame_nominal_wheel_reaction`	413.8 N	Equal-share level-ground weight. Not a rock or slope wheel-load envelope.
Total cutting force `cut_force`	51.4 N	Across four drums at the selected bite and engagement.
Worst modeled contact pressure `ground_pressure`	17.0 kPa	Route pressure estimate, including load transfer and rocker factor.
Structural stress envelope `structure_stress`	109.2 MPa	Envelope of local member stresses; not one assembled-frame FEA result.
Structural deflection envelope `structure_deflection`	0.60 mm	Conservative combination of local/relative deflections.
Upper/lower mount relative displacement `frame_mount_relative_deflection`	0.46 mm	Frame gravity sweep.
Rocker-end flexure `rocker_deflection`	0.59 mm	Local beam-screen displacement.
Suspension-axle flexure `axle_deflection`	0.19 mm	Axle-overhang screen.
Arm-tip flexure `arm_deflection`	0.00584 mm	Relative arm beam displacement.
Loaded CG height `loaded_cg_z`	539.1 mm	Fully loaded travel pose, above level ground.

05 / SCOPE & TRACEABILITY

Model notes & engineering data

Keep assumptions and validity limits attached to the predictions.

Obstacle and structural interpretation

Requested endpoint rock pose is geometrically reachable; this alone does not establish traversal.

Obstacle drum–wheel clearance

Requested-height obstacle clearance screen. Parallel finite cylinders: shared roll is removed, then circular-plane and axial-width gaps are combined. Negative gap means overlap of the ideal solids. A common roll rotation preserves relative clearance in both models; rocker articulation still changes relative geometry.

Ditch requirement remains unverified. Static slope tipping assumes adequate friction; it does not verify climbing. The 5 mm structural limit applies to the modeled envelope of local deflections, not one assembled-frame displacement.

For the plate frame, CF tube stiffness/strength are PROVISIONAL. Connections, dynamic obstacle loads, cutting/recovery frame cases and whole-plate buckling need separate verification.

Standard X44 motors · assumptions

Four wheel motors at 100:1; two drum motors at 40.00:1 (exploratory gearing).

Electrical energy = output-shaft mechanical work / combined efficiency. Drive: 72%; drums: 70%. Both efficiencies are provisional.

Continuous operation unverified. Current-limit inputs of zero mean undecided. Shaft torque-speed capability is enforced separately from energy accounting.

Removed screens: diagnostics only

- `dig_froude_margin`: User removed the uncalibrated speed cap; constant capture/lift response assumes no additional high-speed loss.
- `dump_froude_margin`: User removed the uncalibrated speed cap; actual dumping/retention speed response remains unverified.
- `chassis_clearance_margin`: Legacy filled rectangular envelope does not represent the open plate frame; actual frame/equipment interference remains unverified.

APPENDIX / NATIVE UNITS

Design inputs & output data

141 inputs supplied by file; 0 CLI overrides; 3 remaining inputs use model defaults. Values below retain their native model units.

All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `658422895ded586d418b4ebd64b30c77202e868e2ea838aaab9a3cc69714f7db`.

Input port	Effective value	Native units
wheel_diameter	0.359	m
wheel_width	0.18	m
drum_diameter	0.359	m
drum_length	0.34	m
drum_rpm	120	rpm
dump_rpm	60	rpm
drum_gear_ratio	40	—
cut_depth	0.014	m
arm_wall	0.0015875	m
battery_capacity	259.2	W*h
actuator_speed	0.066	m/s
travel_actuator_length	0.375	m
actuator_retracted_length	0.434	m
actuator_extended_length	0.691	m
top_pivot_height	0.4	m
actuator_pivot_x	0.15	m
bottom_pivot_height	0.18	m
arm_length	0.56	m
pivot_x	0.137	m
wheelbase	0.6	m
track_width	0.57	m
actuator_force_rating	3500	N
arm_box_width	0.0508	m
arm_box_height	0.0508	m
cf_tube_span	0.16	m
cf_tube_wall	0.002	m
plate_spoke_width	0.01875	m
rocker_box_width	0.0508	m
rocker_box_height	0.0508	m
rocker_wall	0.0015875	m

Input port	Effective value	Native units
axle_outer_diameter	0.033	m
axle_wall	0.001	m
rocker_pivot_offset	0.0508	m
empty_slope_degrees	30	deg
loaded_slope_degrees	20	deg
actuator_attachment_fraction	1	—
gravity	9.80665	m/s**2
run_time	900	s
traversal_time_one_way	45	s
travel_speed	1	m/s
acceleration	0.4	m/s**2
turn_angle	3.1415927	rad
overhead_time	8	s
electronics_power	25	W
actuator_idle_power	2	W
soil_density	1500	kg/m**3
berm_density	1500	kg/m**3
fill_fraction	0.9	—
delivery_efficiency	0.9	—
capture_fraction	0.65	—
cut_pressure	18000	Pa
cut_engagement	0.15	—
internal_lift_factor	0.15	—
vertical_cut_fraction	0.5	—
force_imbalance	0.15	—
dump_revolutions	3	—
drum_efficiency	0.7	—
drum_gear_efficiency	0.8	—
drive_efficiency	0.72	—
drive_gear_efficiency	0.8	—
motor_rated_rpm	3000	rpm
traction_coefficient	0.6	—
rocker_load_factor	1.2	—
wheel_deflection_fraction	0.07	—
rolling_base	0.04	—
rolling_pressure_scale	200000	Pa
allowable_ground_pressure	25000	Pa
scrub_coefficient	0.25	—
turn_speed_fraction	0.35	—
turn_accel_fraction	0.5	—
chassis_height	0.35	m

Input port	Effective value	Native units
drum_pair_gap	0.06	m
fixed_mass	6	kg
wheel_areal_mass	8	kg/m**2
drum_areal_mass	5	kg/m**2
motor_mass_per_watt	0.006	kg/W
gear_mass_per_torque	0.05	kg/(N*m)
drive_fixed_mass	0.35	kg
drum_fixed_mass	0.5	kg
battery_specific_energy	140	W*h/kg
battery_usable_fraction	0.8	—
battery_c_rate	6.667	1/h
mass_score_weight	1	—
energy_score_weight	1	—
transport_clearance	0.08	m
mechanism_clearance	0.015	m
minimum_pivot_spacing	0.08	m
minimum_stroke	0.04	m
actuator_housing_allowance	0.04	m
pit_depth	0.45	m
actuator_efficiency	0.65	—
actuator_safety_factor	1.5	—
breakout_force	100	N
actuator_fixed_mass	0.6	kg
actuator_mass_per_force	0.0004	kg/N
actuator_mass_per_power	0.003	kg/W
actuator_mass_per_length	1	kg/m
stored_soil_density	1250	kg/m**3
usable_volume_fraction	0.9	—
scoop_opening	0.035	m
scoop_depth_fraction	0.5	—
scoops_per_revolution	4	—
max_dig_froude	0.3	—
max_dump_froude	0.2	—
drum_spinup_time	2	s
drum_torque_service_factor	1.25	—
minimum_dig_depth	0.005	m
minimum_payload	0.05	kg
fresh_site_distance	0.8	m
site_speed	0.1	m/s
available_excavation_area	12	m**2
tube_density	2700	kg/m**3

Input port	Effective value	Native units
tube_youngs_modulus	6.9e+10	Pa
tube_yield_strength	2.4e+08	Pa
structure_safety_factor	2	—
structure_load_factor	1.5	—
allowable_deflection	0.005	m
joint_mass	2	kg
differential_mass	1	kg
obstacle_height	0.4	m
slope_angle	0.52359878	rad
frame_tube_diameter	0.055	m
frame_wall	0.002	m
arm_tube_diameter	0.05	m
plate_thickness	0.00635	m
cf_tube_inner_diameter	0.0254	m
wheel_rocker_gap	0.0127	m
plate_hub_radius	0.03	m
plate_cutout_clearance	0.005	m
frame_mount_clearance	0.003	m
minimum_axle_overhang	0.01	m
plate_buckling_k	1	—
cf_density	1600	kg/m**3
cf_flexural_modulus	7e+10	Pa
cf_flexural_strength	3.5e+08	Pa
drive_motor_power	835.4	W
drum_motor_power	835.4	W
drive_gear_ratio	100	—
motor_bus_voltage	12	V
x44_motor_mass	0.34019428	kg
drive_gearbox_mass	0.697	kg
drum_gearbox_mass	0.753	kg
drive_stator_current_limit	0	A
drum_stator_current_limit	0	A

Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

Port	Lower	Evaluated	Upper	Native units
wheel_diameter	0.18	0.359	0.359	m
wheel_width	0.05	0.18	0.359	m
drum_diameter	0.16	0.359	0.359	m
drum_length	0.12	0.34	0.359	m
drum_rpm	5	120	120	rpm
dump_rpm	5	60	60	rpm
drum_gear_ratio	9	40	125	—
cut_depth	0.003	0.014	0.025	m
arm_wall	0.001	0.0015875	0.005	m
battery_capacity	10	259.2	300	W*h
actuator_speed	0.005	0.066	0.08	m/s
travel_actuator_length	0.3	0.375	0.65	m
actuator_retracted_length	0.25	0.434	0.65	m
actuator_extended_length	0.45	0.691	1.05	m
top_pivot_height	0.35	0.4	0.7	m
actuator_pivot_x	0.05	0.15	0.3	m
bottom_pivot_height	0.09	0.18	0.3	m
arm_length	0.3	0.56	0.7	m
pivot_x	0.05	0.137	0.3	m
wheelbase	0.38	0.6	0.65	m
track_width	0.5	0.57	0.699	m
actuator_force_rating	500	3500	6000	N
arm_box_width	0.025	0.0508	0.1	m
arm_box_height	0.025	0.0508	0.1	m
cf_tube_span	0.16	0.16	0.46	m
cf_tube_wall	0.001	0.002	0.006	m
plate_spoke_width	0.012	0.01875	0.07	m
rocker_box_width	0.02	0.0508	0.065	m
rocker_box_height	0.025	0.0508	0.09	m
rocker_wall	0.001	0.0015875	0.005	m
axle_outer_diameter	0.025	0.033	0.09	m
axle_wall	0.001	0.001	0.006	m
rocker_pivot_offset	0.02	0.0508	0.15	m

All output ports

Output port	Value	Native units
dry_mass	52.679149	kg
payload_mass	116.12008	kg
wheel_mass	12.974577	kg
drum_mass	13.904561	kg
drive_mass	4.1487771	kg
battery_mass	1.8514286	kg
rover_length	1.4733605	m
rover_width	0.75	m
rover_height	0.72970071	m
ground_pressure	17013.242	Pa
actual_travel_speed	0.57462968	m/s
dig_time	24.339721	s
travel_time	90	s
dump_time	10	s
turn_time	4.308154	s
cycle_time	159.63929	s
cycle_energy	6.6779735	W*h
cycles	5.6377098	—
berm_volume	0.39279079	m**3
run_energy	37.648476	W*h
average_power	150.5939	W
peak_power	1204.6755	W
mass_points	2187.1772	—
energy_points	1043.3113	—
construction_points	3230.4884	—
no_load_speed	1.4582122	m/s
loaded_motor_speed_limit	1.1278696	m/s
loaded_power_speed_limit	7.0870032	m/s
stall_drive_force	7333.0043	N
actuation_time	9.5757576	s
arm_energy	1742.1554	J
arm_peak_power	459.59786	W
required_actuator_force	2243.4743	N
linkage_mass	8.4593058	kg
stroke	0.316	m
stowed_x	0.55718024	m
stowed_y	0.55020071	m

Output port	Value	Native units
travel_x	0.41312165	m
travel_y	0.66719281	m
extended_x	0.61583877	m
extended_y	-0.11036775	m
sweep_x_min	0.41312165	m
sweep_x_max	0.697	m
actuator_leverage_max	3.0322552	—
actuator_attachment_distance	0.56	m
chassis_clearance_margin	-0.086979361	m
stowed_length	1.4733605	m
stowed_height	0.72970071	m
deployed_length	1.7529859	m
deployed_height	0.84669281	m
route_distance	18	m
route_turn_angle	360	deg
route_translation_time	64.857122	s
route_turn_time	25.142878	s
route_peak_acceleration	0.13207971	m/s**2
payload_volume	0.092896066	m**3
fresh_volume	0.11909752	m**3
storage_capacity	0.094791587	m**3
reachable_soil_mass	116.12008	kg
target_payload_mass	118.48948	kg
trench_depth	0.28986775	m
excavation_footprint	1.1771014	m**2
cut_force	51.408	N
drum_froude	2.890437	—
dig_overlap_time	4.7878788	s
dig_power	720.07763	W
dump_power	139.60543	W
dig_energy	10418.748	J
dump_energy	550.95337	J
drum_power_margin	710.06659	W
drum_torque_margin	50.505831	N*m
dump_torque_margin	71.742334	N*m
dig_froude_margin	-2.590437	—
dump_froude_margin	-0.52260925	—
penetration_margin	490.90198	N
dig_traction_margin	286.82999	N
cut_depth_margin	0.0035	m
dig_reach_margin	0.28486775	m

Output port	Value	Native units
payload_margin	116.07008	kg
structure_mass	5.3405005	kg
rocker_mass	1.0269115	kg
arm_linear_mass	0.84374831	kg/m
frame_section_margin	0.00725	m
arm_section_margin	0.0111125	m
structure_stress	1.091514e+08	Pa
structure_stress_margin	10848601	Pa
structure_deflection	0.00059998122	m
structure_deflection_margin	0.0044000188	m
structure_buckling_margin	1436.1811	N
empty_yaw_inertia	7.479752	kg*m**2
empty_cg_x	0	m
empty_cg_z	0.38851406	m
loaded_yaw_inertia	33.996689	kg*m**2
loaded_cg_x	0	m
loaded_cg_z	0.53906183	m
half_yaw_inertia	20.738221	kg*m**2
half_cg_x	0.21659775	m
half_cg_z	0.50325369	m
loaded_sprung_mass	150.64897	kg
loaded_sprung_cg_x	0	m
loaded_sprung_cg_z	0.58220888	m
relocation_time	18.203539	s
relocation_energy	280.4598	J
relocation_peak_power	22.868899	W
fresh_site_margin	0.06	m
site_drive_margin	202.90751	N
site_turn_time	9.9535391	s
mobility_energy	7057.4054	J
drive_peak_power	304.4148	W
turn_peak_power	149.47891	W
ground_pressure_margin	7986.7576	Pa
straight_motor_margin	3502.7698	N
turn_motor_margin	5281.8947	N
straight_traction_margin	238.48552	N
turn_traction_margin	-112.61802	N
straight_power_margin	2459.8504	W
turn_power_margin	2571.4042	W
mobility_stability_margin	0.26651053	m
trench_traverse_margin	0.069132254	m

Output port	Value	Native units
tip_stability_margin	-0.022881399	m
tip_worst_mask	3	—
obstacle_pose_margin	0.14975636	m
obstacle_tip_margin	0.14644954	m
obstacle_pose_valid	1	—
obstacle_evaluated_height	0.4	m
obstacle_worst_wheel	0	—
obstacle_chassis_pitch	0.39141122	rad
obstacle_chassis_roll	0.33244614	rad
obstacle_loaded_cg_x	-0.11794371	m
obstacle_loaded_cg_y	-0.093266642	m
obstacle_loaded_cg_z	0.59066434	m
obstacle_drum_wheel_gap_bound	0.02831708	m
obstacle_drum_wheel_margin	0.01331708	m
obstacle_clearance_pose_valid	1	—
obstacle_clearance_evaluated_height	0.4	m
slope_pitch_tip_margin	-0.011227492	m
slope_roll_tip_margin	-0.026227492	m
slope_tip_margin	-0.026227492	m
slope_tip_angle	0.48634024	rad
rock_sweep_tip_margin	0.14644954	m
rock_sweep_worst_height	0.4	m
rock_sweep_raised_wheel	0	—
rock_sweep_pitch	0.39141122	rad
rock_sweep_roll	0.33244614	rad
rock_sweep_cg_x	-0.11794371	m
rock_sweep_cg_y	-0.093266642	m
rock_sweep_cg_z	0.59066434	m
rock_sweep_wheel_load_margin	-48.364016	N
rock_sweep_load_height	0.4	m
rock_sweep_load_raised_wheel	2	—
rock_sweep_unloaded_wheel	2	—
rock_sweep_pose_valid	1	—
rock_sweep_evaluated_height	0.4	m
rock_sweep_samples	41	—
rock_sweep_force_balance_error	3.126388e-13	N
empty_slope_slip	1	—
empty_slope_solution_valid	0	—
empty_slope_traversal_margin	-0.33988069	—
empty_slope_force_shortfall	38.015137	N
empty_slope_sinkage	0.0068503147	m

Output port	Value	Native units
empty_slope_transfer_flag_count	0	—
loaded_slope_slip	1	—
loaded_slope_solution_valid	0	—
loaded_slope_traversal_margin	-0.12250103	—
loaded_slope_force_shortfall	47.63835	N
loaded_slope_sinkage	0.015877933	m
loaded_slope_transfer_flag_count	1	—
plate_mass_total	1.0521221	kg
cf_tube_mass_total	0.1762911	kg
suspension_axle_mass	0.085175865	kg
suspension_bearing_span	0.3138	m
cf_tube_outer_diameter	0.0294	m
plate_geometry_margin	0.037570066	m
frame_packing_margin	0.063725	m
plate_geometry_valid	1	—
structure_cg_z	0.24042695	m
structure_sprung_cg_z	0.24888465	m
structure_yaw_inertia	0.078448908	kg*m**2
plate_stress	16590191	Pa
plate_stress_margin	1.0340981e+08	Pa
plate_buckling_margin	1436.1811	N
cf_bending_stress	1.0536889e+08	Pa
cf_bending_margin	69631108	Pa
frame_mount_relative_deflection	0.00045590901	m
rocker_deflection	0.00059120088	m
axle_deflection	0.00018898313	m
arm_deflection	5.8372827e-06	m
rocker_buckling_margin	116008.09	N
arm_buckling_margin	31673.333	N
frame_gravity_actuator_force	2910.3732	N
frame_nominal_wheel_reaction	413.83875	N
frame_force_balance_error	2.0463631e-12	N
frame_load_solution_valid	1	—